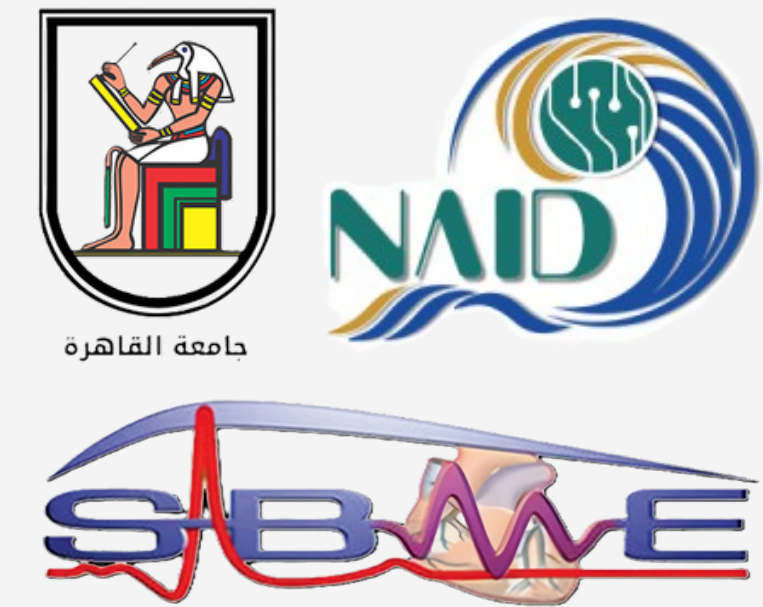


AUTISPHERE: A MR & VR REHABILITATION PLATFORM FOR AUTISM TO ENHANCE DAILY LIVING AND EMOTIONAL SKILLS

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01. INTRODUCTION

Autistic children in Egypt struggle with emotion recognition and daily living skills. Traditional therapies are costly, inconsistent, and hard to generalize.

Families face:

- Shortages of therapists
- Long waitlists
- Limited progress tracking

While VR has potential, it lacks Arabic localization and therapist analytics, creating barriers for wider use.

02. OBJECTIVE

Develop a Mixed Reality (MR) rehabilitation platform to support autistic learners (ages 6–18) in practicing social-communication and daily living skills in safe, culturally relevant environments.

03. METHODOLOGY

Game Scenarios:

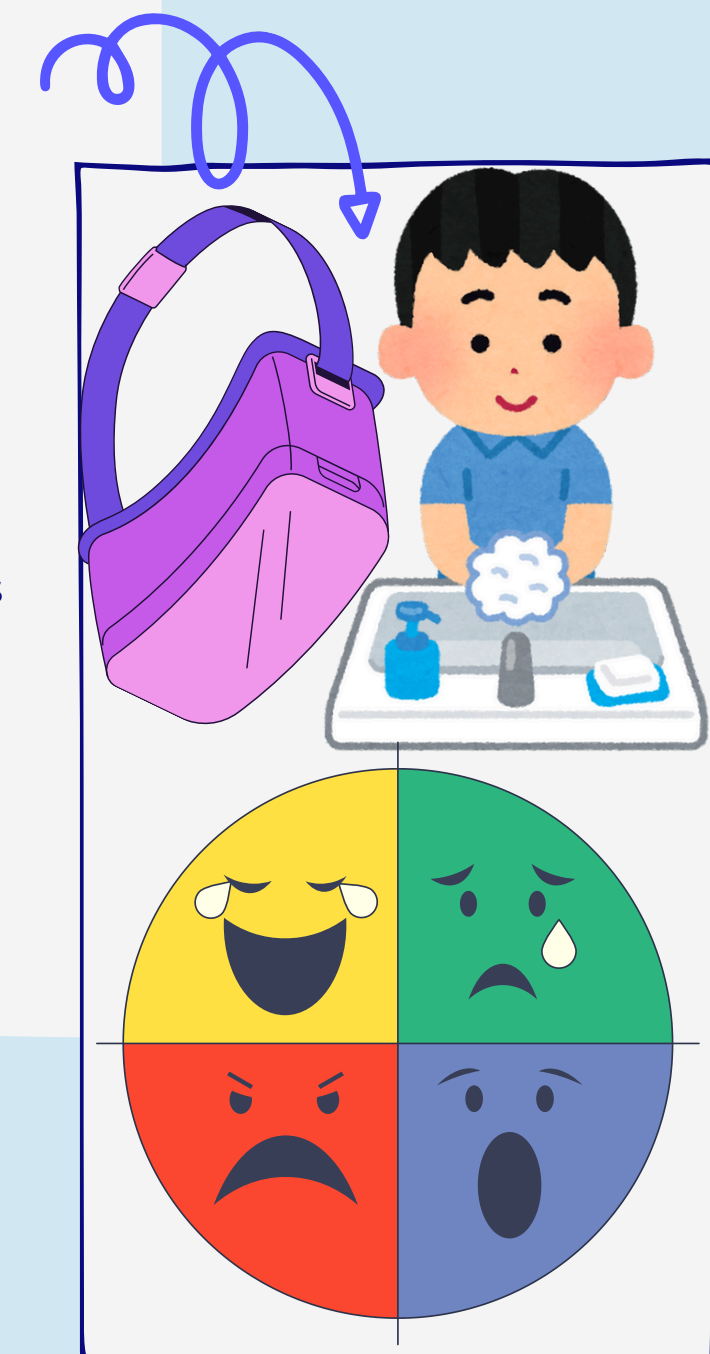
- Handwashing (self-care)
- Emotion recognition (social cues)
- Supermarket/mall navigation (community living)

Design Approach:

- Built in Unity with Quest 3 and HoloLens 2
- Autism-friendly UI: simple graphics, consistent colors, minimal inputs
- Emotion recognition: 3D avatars with clear facial expressions
- Navigation: teleport or smooth walking with comfort focus

Accessibility:

- Gentle sounds, calm voice narration
- No flashing/flicker, uncluttered visuals
- Tasks at child's pace, repeatable prompts



Related Literature

- MR is more immersive than AR and safer than VR (less cybersickness).
- 112 XR autism interventions (2015–2024) → most VR/AR, almost none MR.
- MR studies show ↑ social initiation, ↑ gait speed, and 82% headset usability mastery.
- Gap: No MR tools localized in Arabic or tailored to Egyptian clinical needs.

04. RESULTS/FINDINGS

Clinical/Educational:

- Improved emotion recognition
- Increased independence in daily living skills
- Better transfer of skills to school and home settings

Operational:

- Standardized lesson delivery across therapists
- Automatic session tracking reduces therapist workload

05. ANALYSIS

Equity & Access:

- Arabic localization improves inclusivity
- Offline-first design ensures usability in rural areas

Economic Value:

- Lower cost per mastered skill compared to traditional therapy
- Collected data strengthens opportunities for funding and corporate sponsorships (CSR initiatives)

06. CONCLUSION

AutiSphere = Scalable, affordable, clinically relevant MR tool.

- Empowers autistic individuals with safe, independent training
- First high-quality Arabic MR solution for autism therapy in Egypt
- Future: expand to dressing, shopping, transport, and AI-based personalization

SCAN ME

